EXPERIMENT NO 14

Determination and plotting of Polarization in different Dielectric Medium with their relative permittivity and to find electric susceptibility.

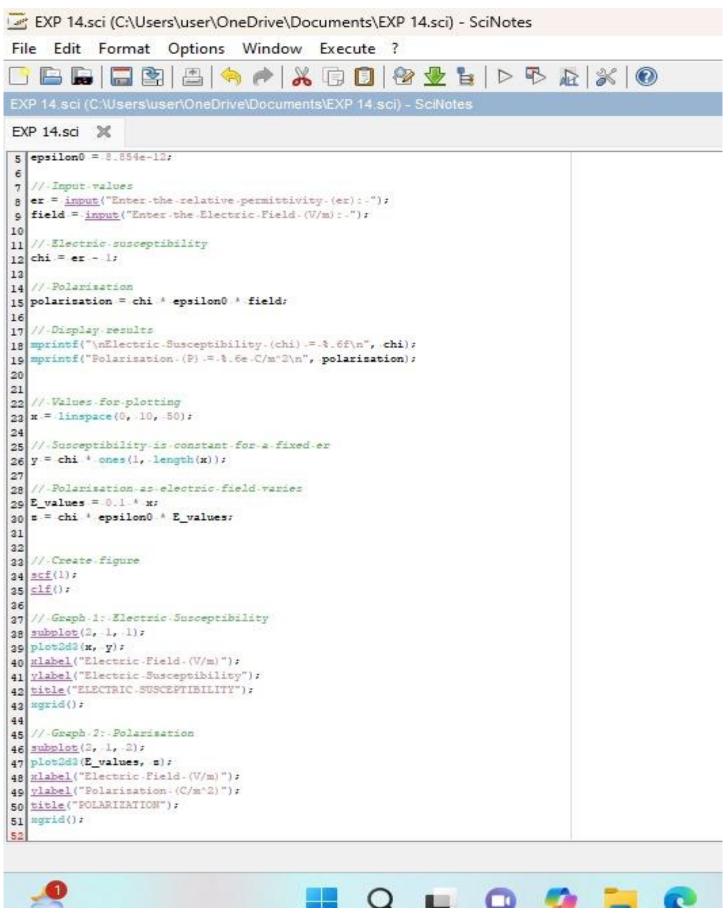
Aim:

To determine and plot polarization in different dielectric medium with their relative permittivity and to find electric susceptibility.

Software required:

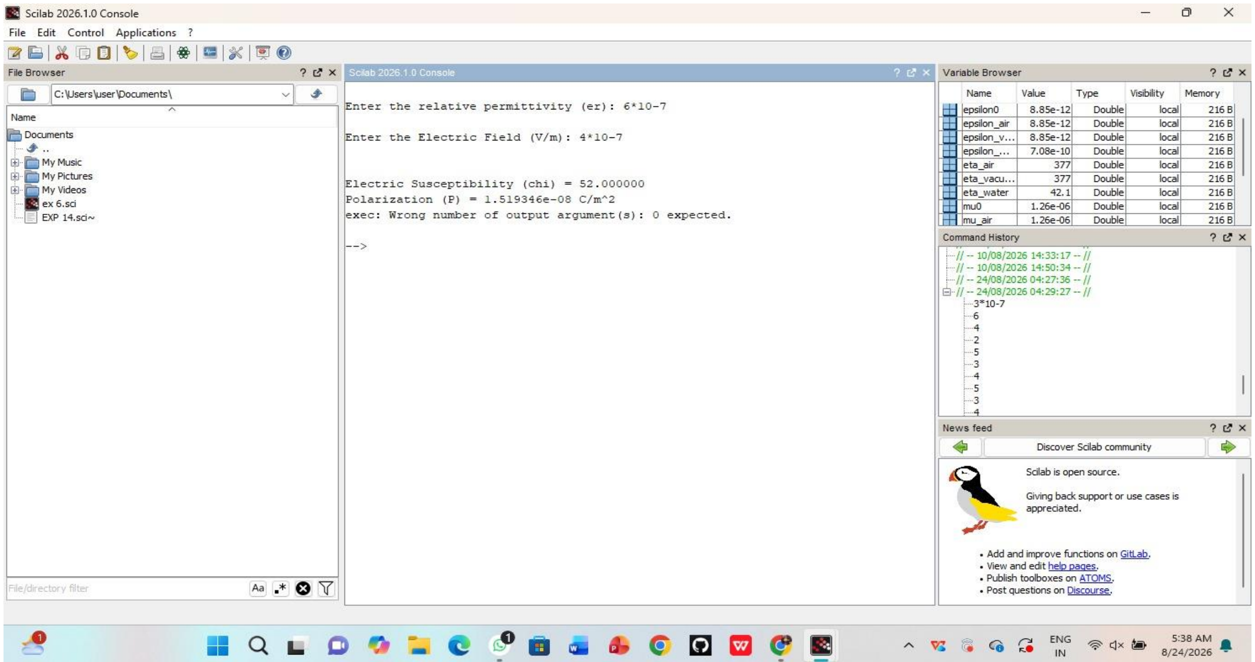
- SCILAB version 6.0.1

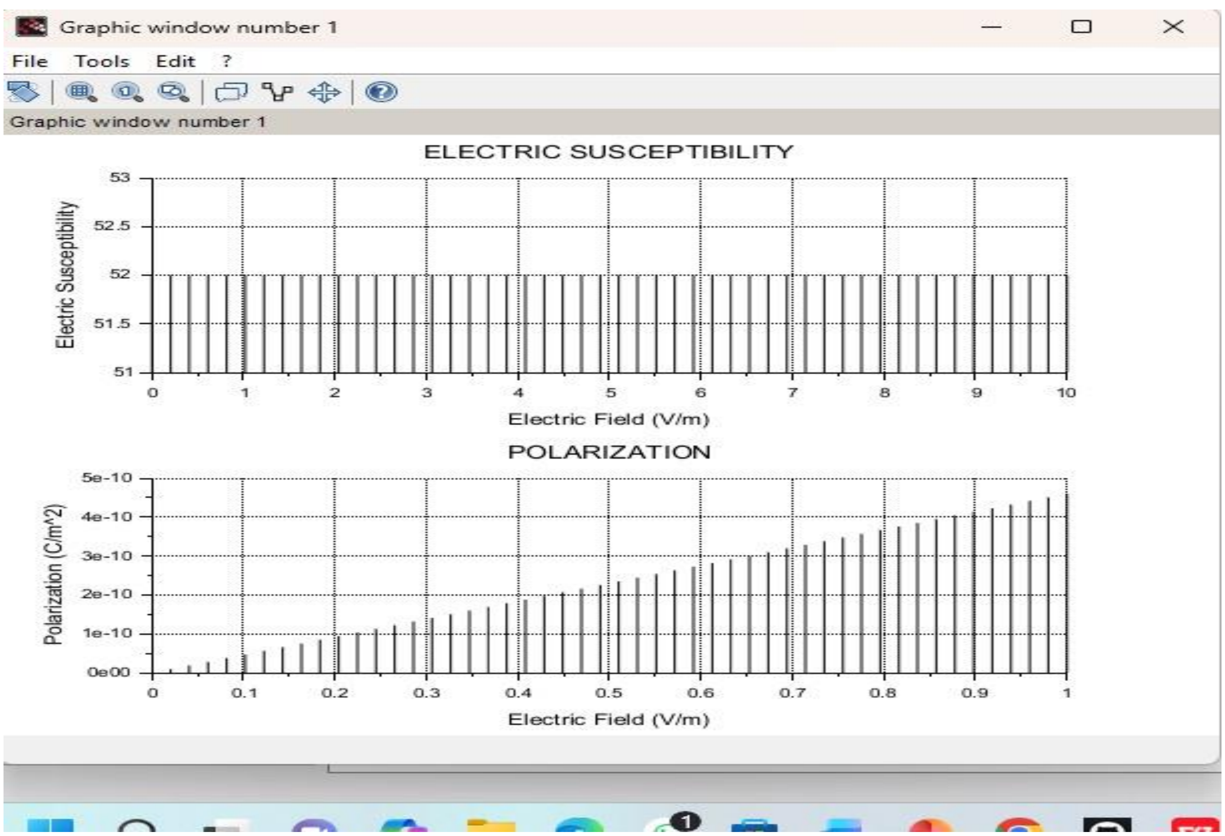
Program:




```
epsilon0 = 8.854e-12;

// Input values
er = input("Enter the relative permittivity (er): ");
field = input("Enter the Electric Field (V/m): ");

// Electric susceptibility
chi = er - 1;

// Polarization
polarization = chi * epsilon0 * field;

// Display results
mprintf("\nElectric Susceptibility (chi) = %.6f\n", chi);
mprintf("Polarization (P) = %.6e C/m^2\n", polarization);


// Values for plotting
x = linspace(0, 10, 50);

// Susceptibility is constant for a fixed er
y = chi * ones(1, length(x));

// Polarization as electric field varies
E_values = 0.1 * x;
z = chi * epsilon0 * E_values;


// Create figure
scf(1);
clf();

// Graph 1: Electric Susceptibility
subplot(2, 1, 1);
plot2d3(x, y);
xlabel("Electric Field (V/m)");
ylabel("Electric Susceptibility");
title("ELECTRIC SUSCEPTIBILITY");
xgrid();

// Graph 2: Polarisation
subplot(2, 1, 2);
plot2d3(E_values, z);
xlabel("Electric Field (V/m)");
ylabel("Polarisation (C/m^2)");
title("POLARIZATION");
xgrid();
```

Output:


Scilab 2026.1.0 Console
Enter the relative permittivity (er): 6*10-7
Enter the Electric Field (V/m): 4*10-7
Electric Susceptibility (chi) = 52.000000
Polarization (P) = 1.519346e-08 C/m^2
exec: Wrong number of output argument(s): 0 expected.
-->


Graphic window number 1
ELECTRIC SUSCEPTIBILITY
Electric Susceptibility
Electric Field (V/m)
POLARIZATION
Polarization (C/m^2)
Electric Field (V/m)

Theoretical calculations:

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Output:

Enter the ε_r value = 6×10^{-7}

Enter the Electric Field = 4×10^{-7}

Case 3:

Enter the value of mr = 5×10^{-7}

Enter the value of H = 5 cm

Enter the value of length = 2 cm

Enter the value of width = 3 mm

Practical Output:

" Electric susceptibility (chi) = 52.000000 "

" Polarization (P) = 1.51934e - C/m^2 "

Theoretical Calculation:

Relative permittivity, $\varepsilon_r = 53$

Electric field, $E = 4 \times 10^{7}$ V/m

Permittivity of free space,

$\varepsilon_0 = 8.854 \times 10^{-12}$ F/m

(1) Electric Susceptibility

$\chi_E = 53 - 1$

$\chi_E = 52$

(2) Polarization

$P = \varepsilon_0 \chi_E E$

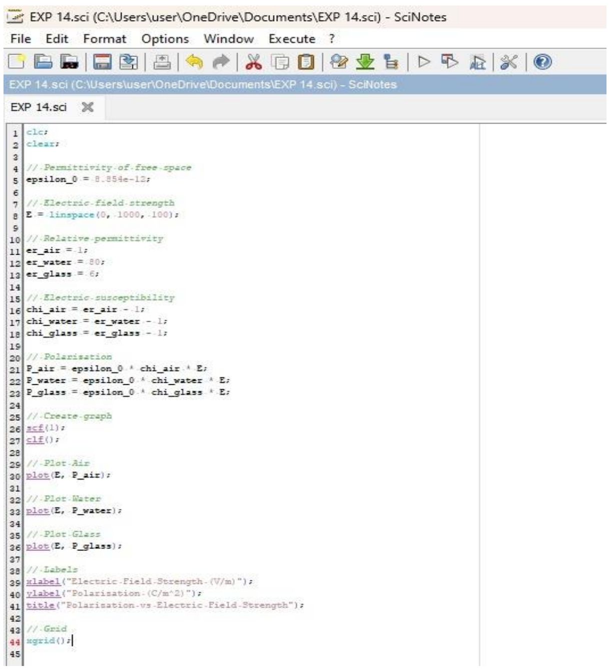
$P = 8.854 \times 10^{-12} \times 52 \times 4 \times 10^{7}$

$P = 1.839 \times 10^{-2}$ C/m2

$P = 1.839632 \times 10^{-2}$ C/m2

Case 1: To determine and plot the polarization and electric susceptibility of electromagnetic waves in three dielectric media: Air, Water, and Glass, using their relative permittivity.

Program:

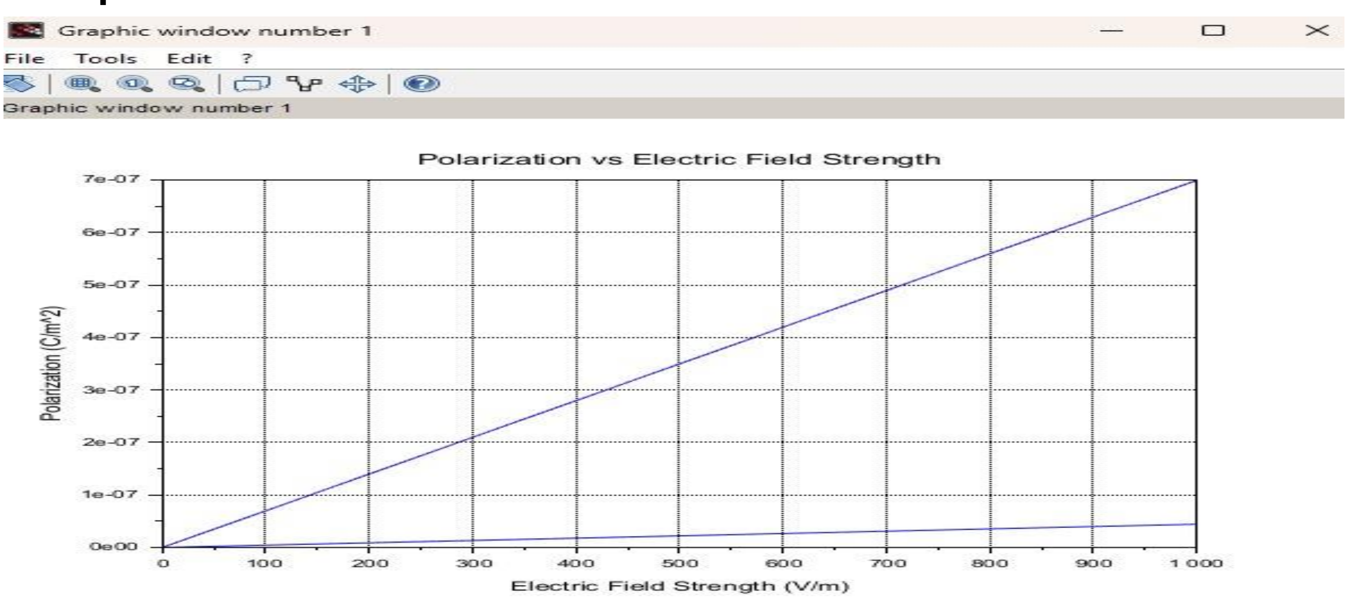

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?

EXP 14.sci

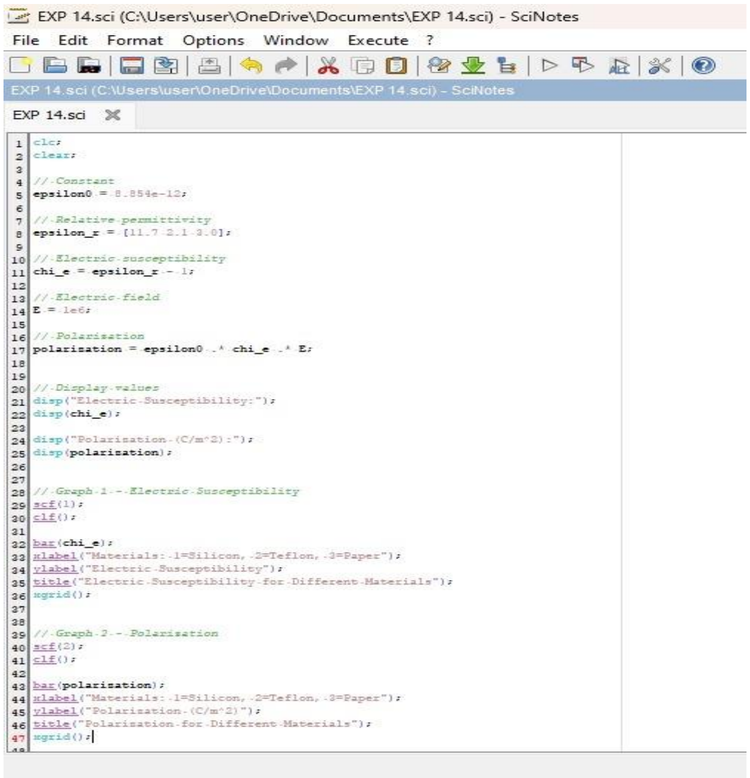
```
clc;
clear;

// Permittivity of free space
epsilon_0 = 8.854e-12;

// Electric field strength
E = linspace(0, 1000, 100);

// Relative permittivity
er_air = 1;
er_water = 80;
er_glass = 6;

// Electric susceptibility
chi_air = er_air - 1;
chi_water = er_water - 1;
chi_glass = er_glass - 1;

// Polarization
P_air = epsilon_0 * chi_air * E;
P_water = epsilon_0 * chi_water * E;
P_glass = epsilon_0 * chi_glass * E;

// Create graph
scf(1);
clf();

// Plot Air
plot(E, P_air);

// Plot Water
plot(E, P_water);

// Plot Glass
plot(E, P_glass);

// Labels
xlabel("Electric Field Strength (V/m)");
ylabel("Polarisation (C/m^2)");
title("Polarisation vs Electric Field Strength");

// Grid
xgrid();
```

Output:



Case 2: To calculate and plot the polarization and electric susceptibility in Silicon, Teflon, and Paper, and understand how their relative permittivity affects the polarization.

Program:

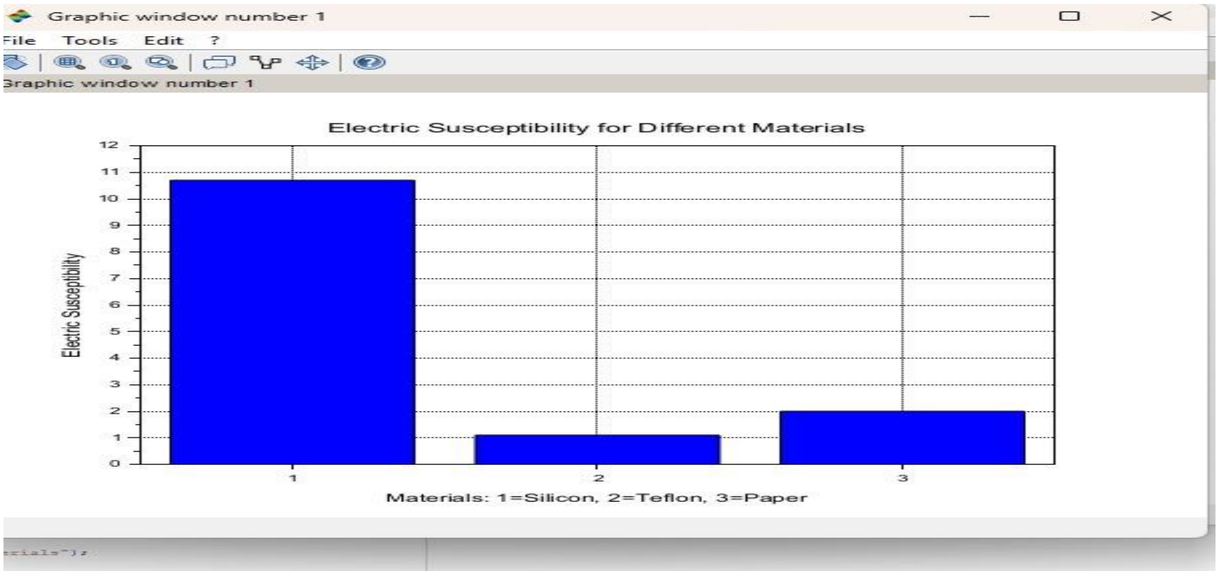

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?

EXP 14.sci

```
clc;
clear;

// Constant
epsilon0 = 8.854e-12;

// Relative permittivity
epsilon_r = [11.7 2.1 3.0];

// Electric susceptibility
chi_e = epsilon_r - 1;

// Electric field
E = 1e6;

// Polarization
polarization = epsilon0 .* chi_e .* E;


// Display values
disp("Electric Susceptibility:");
disp(chi_e);

disp("Polarization (C/m^2):");
disp(polarization);


// Graph 1 - Electric Susceptibility
scf(1);
clf();

bar(chi_e);
xlabel("Materials: 1=Silicon, 2=Teflon, 3=Paper");
ylabel("Electric Susceptibility");
title("Electric Susceptibility for Different Materials");
xgrid();


// Graph 2 - Polarization
scf(2);
clf();

bar(polarization);
xlabel("Materials: 1=Silicon, 2=Teflon, 3=Paper");
ylabel("Polarization (C/m^2)");
title("Polarization for Different Materials");
xgrid();
```

Output:



Case 3: To explore how different dielectric materials (Rubber, Oil, and Plastic) affect the polarization and electric susceptibility, and to compare their relative permitivity.

Program:

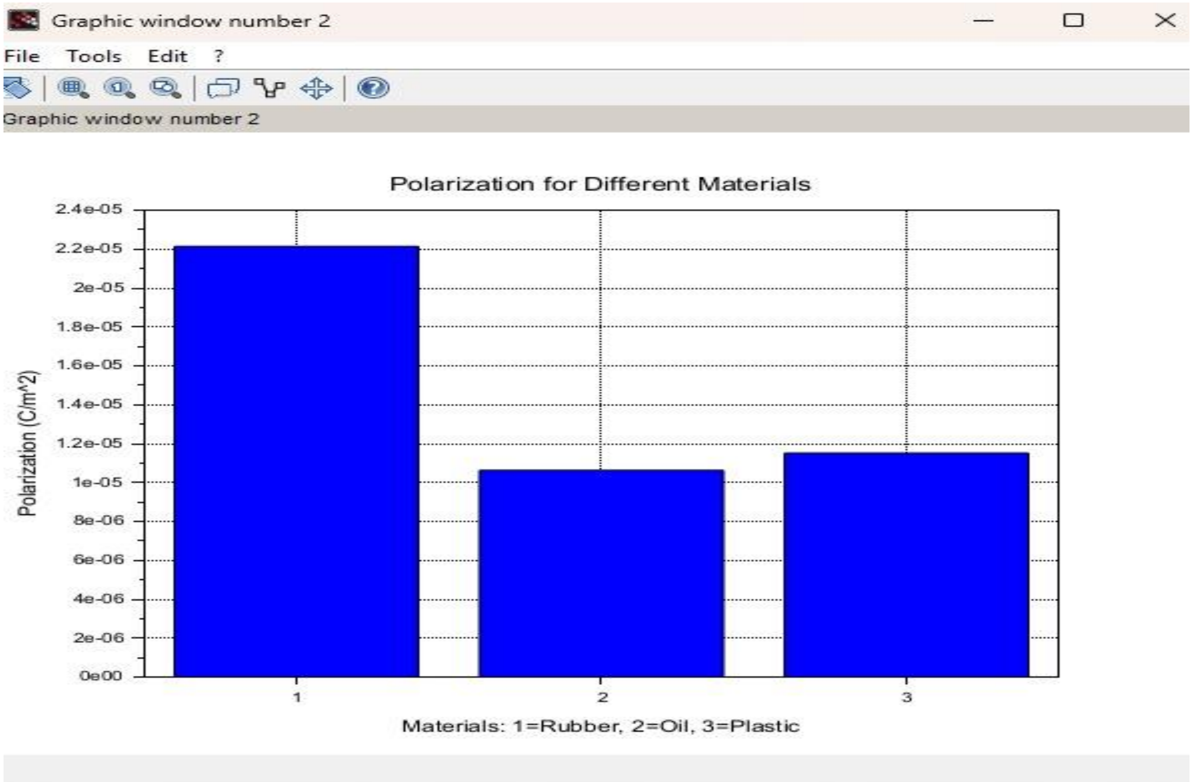
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?

EXP 14.sci

```
clc;
clear;

// Constant
epsilon0 = 8.854e-12;

// Relative permittivity
epsilon_r = [3.5 2.2 2.3];

// Electric susceptibility
chi_e = epsilon_r - 1;

// Applied electric field
E = 1e6;

// Polarization
polarization = epsilon0 .* chi_e .* E;


// ====================================
// DISPLAY RESULTS
// ====================================

disp("Electric Susceptibility:");
disp(chi_e);

disp("Polarization (C/m^2):");
disp(polarization);


// ====================================
// GRAPH 1: ELECTRIC SUSCEPTIBILITY
// ====================================

scf(1);
clf();

bar(chi_e);

xlabel("Materials: 1=Rubber, 2=Oil, 3=Plastic");
ylabel("Electric Susceptibility");
title("Electric Susceptibility for Different Materials");

xgrid();

// ====================================
// GRAPH 2: POLARIZATION
// ====================================

scf(2);
clf();

bar(polarization);

xlabel("Materials: 1=Rubber, 2=Oil, 3=Plastic");
ylabel("Polarization (C/m^2)");
title("Polarization for Different Materials");

xgrid();
```

Output:



Case 1: Examine the effect of length of a ferromagnetic material on the magnetic flux density (B) for the

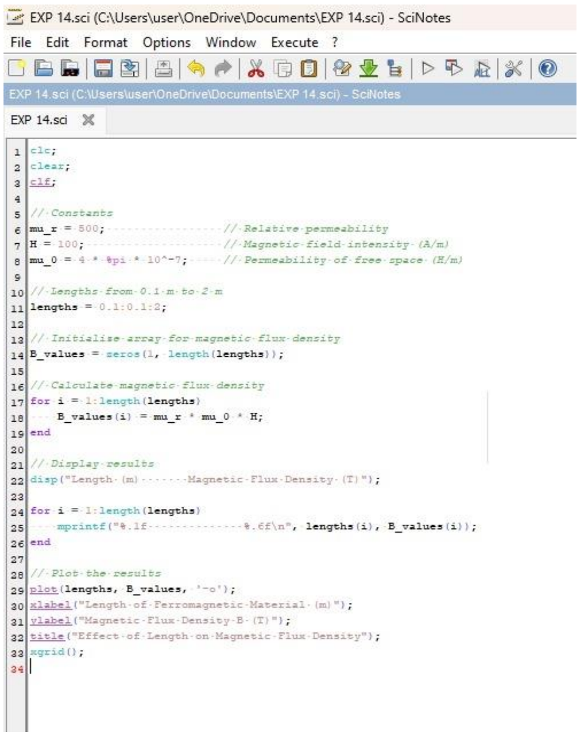
following conditions

 The material is ferromagnetic with relative permeability (μr) of 500.

 The magnetic field intensity (H) is held constant at 100 A/m.

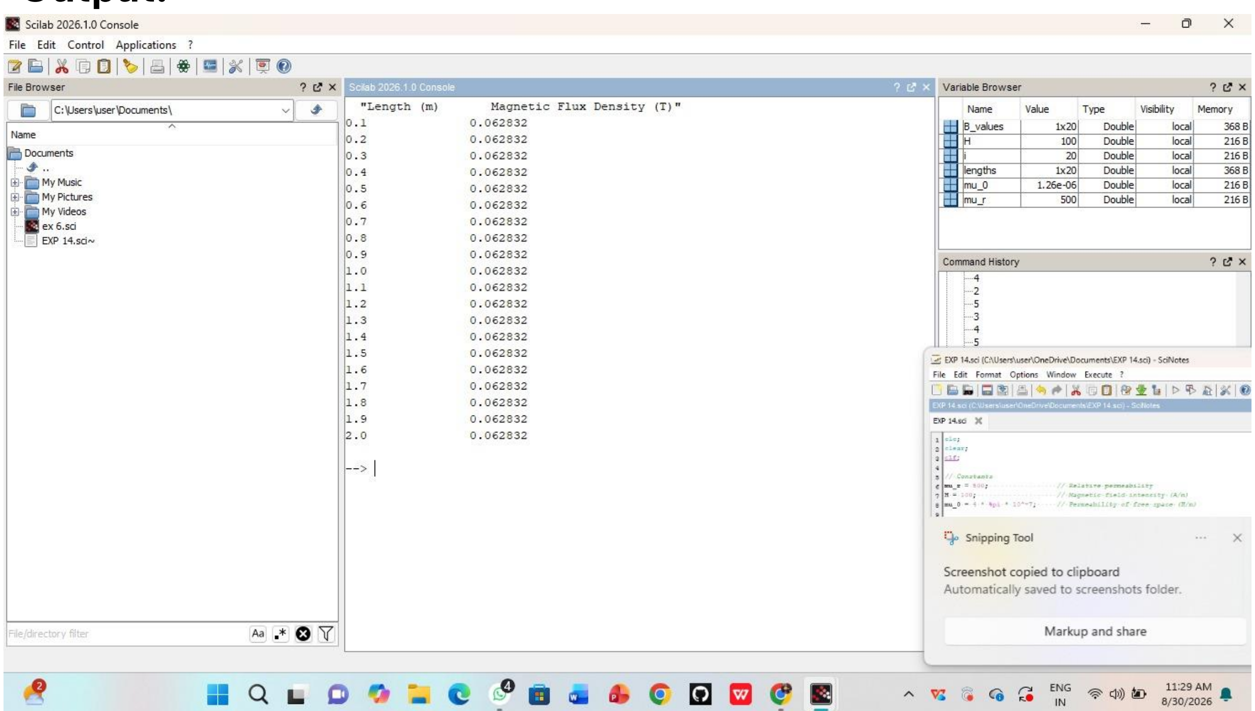
 The length of the material varies from 0.1 m to 2 m.

Program:



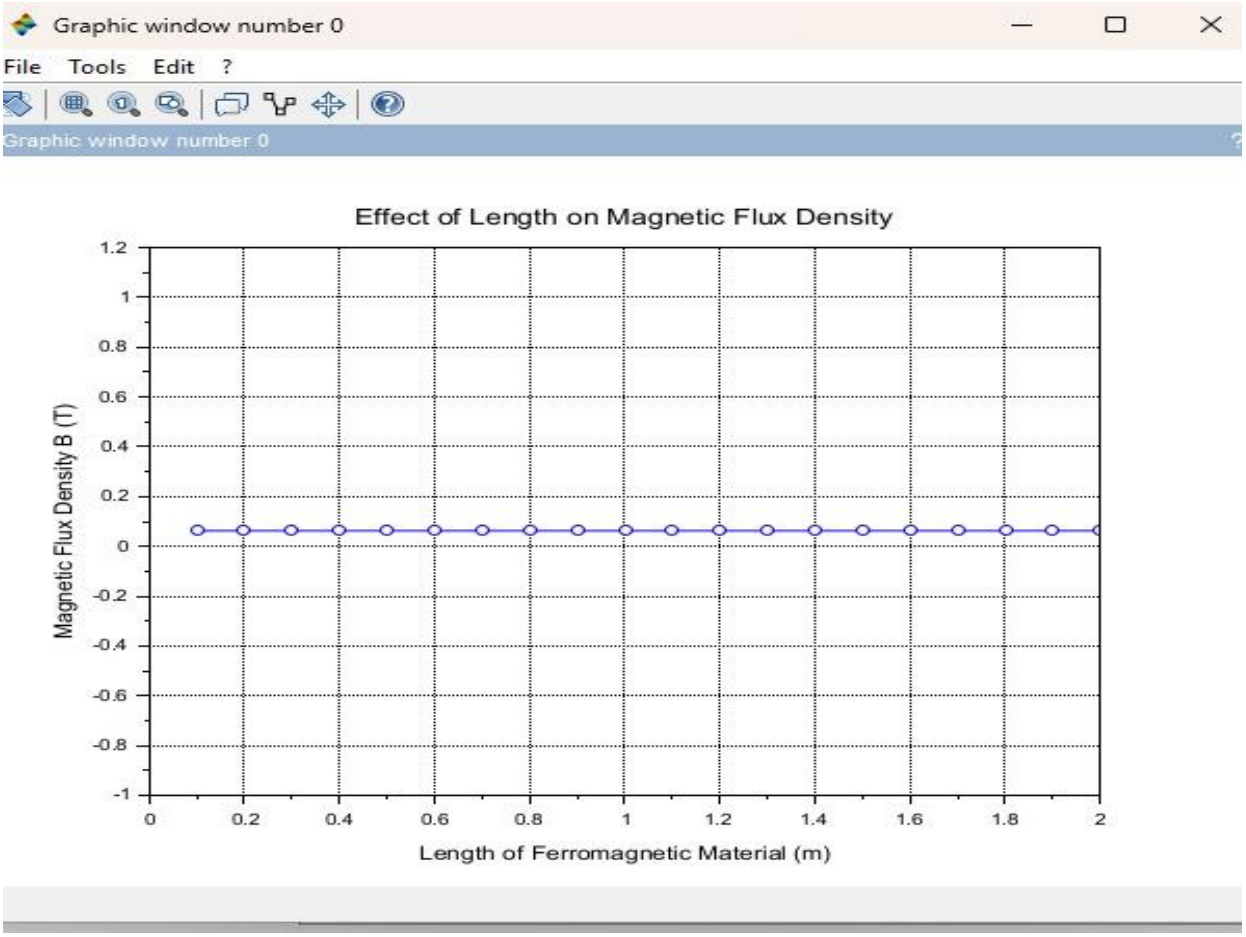

```
clc;
clear;
clf;

// Constants
mu_r = 500;                  // Relative permeability
H = 100;                     // Magnetic field intensity (A/m)
mu_0 = 4 * %pi * 10^-7;      // Permeability of free space (H/m)

// Lengths from 0.1 m to 2 m
lengths = 0.1:0.1:2;

// Initialise array for magnetic flux density
B_values = zeros(1, length(lengths));

// Calculate magnetic flux density
for i = 1:length(lengths)
    B_values(i) = mu_r * mu_0 * H;
end

// Display results
disp("Length (m)       Magnetic Flux Density (T)");

for i = 1:length(lengths)
    mprintf("%.1f              %.6f\n", lengths(i), B_values(i));
end

// Plot the results
plot(lengths, B_values, '-o');
xlabel("Length of Ferromagnetic Material (m)");
ylabel("Magnetic Flux Density B (T)");
title("Effect of Length on Magnetic Flux Density");
xgrid();
```

Output:




```
 "Length (m)       Magnetic Flux Density (T)"
0.1              0.062832
0.2              0.062832
0.3              0.062832
0.4              0.062832
0.5              0.062832
0.6              0.062832
0.7              0.062832
0.8              0.062832
0.9              0.062832
1.0              0.062832
1.1              0.062832
1.2              0.062832
1.3              0.062832
1.4              0.062832
1.5              0.062832
1.6              0.062832
1.7              0.062832
1.8              0.062832
1.9              0.062832
2.0              0.062832

-->
```

Name	Value	Type	Visibility	Memory
B_values	1x20	Double	local	368 B
H	100	Double	local	216 B
i	20	Double	local	216 B
lengths	1x20	Double	local	368 B
mu_0	1.26e-06	Double	local	216 B
mu_r	500	Double	local	216 B



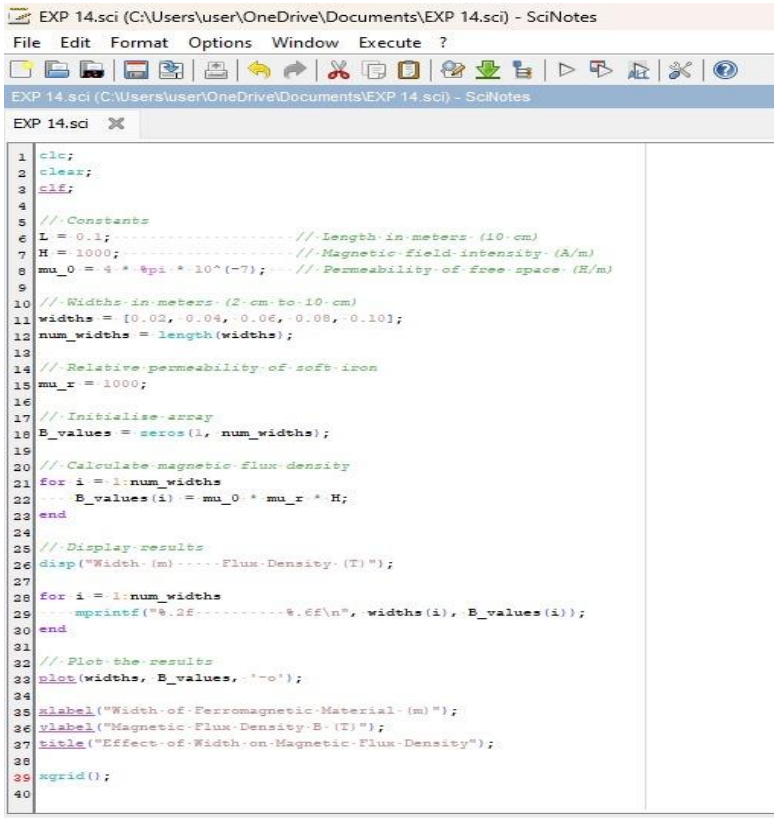
Case 2: To investigate how the magnetic flux density (B) in a ferromagnetic material changes with

different widths at a constant length and magnetic field intensity.

Conditions:

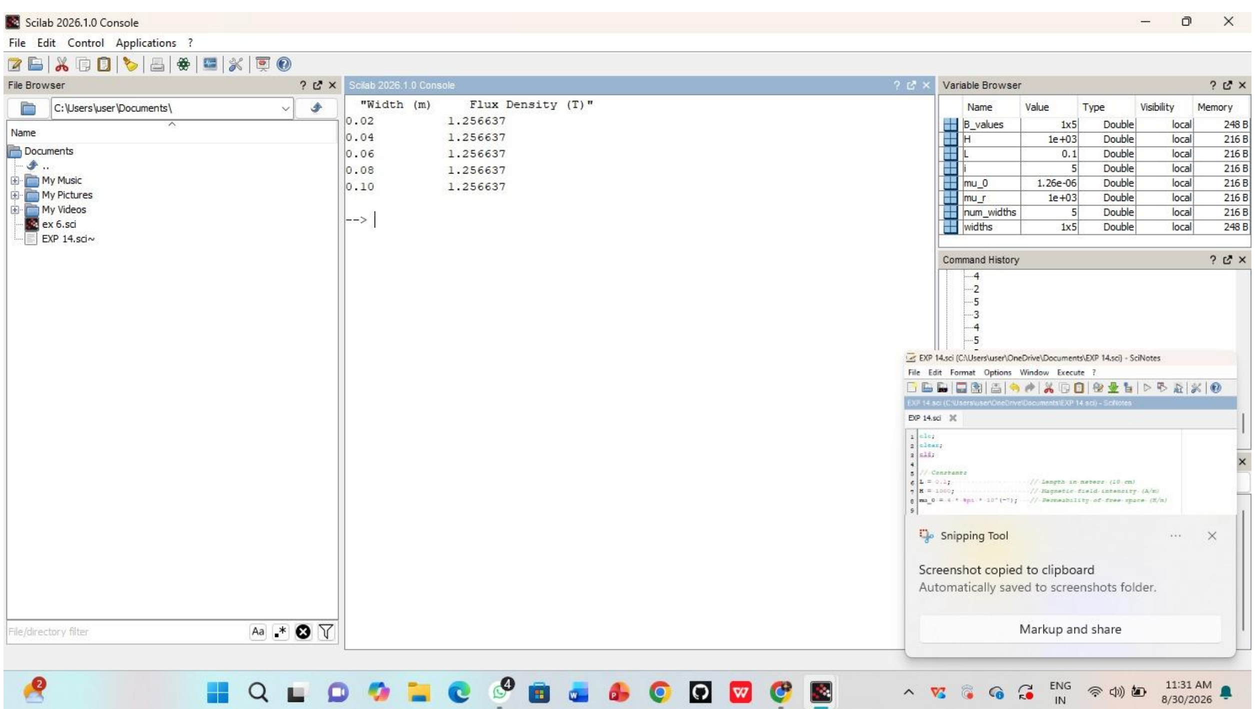
- Material: Soft iron (same as in Case Study 1)
- Length (L): Constant at 10 cm
- Magnetic Field Intensity (H): Constant at 1000 A/m
- Width (W): Varies from 2 cm to 10 cm (i.e., 2 cm, 4 cm, 6 cm, 8 cm, 10 cm)

Program:

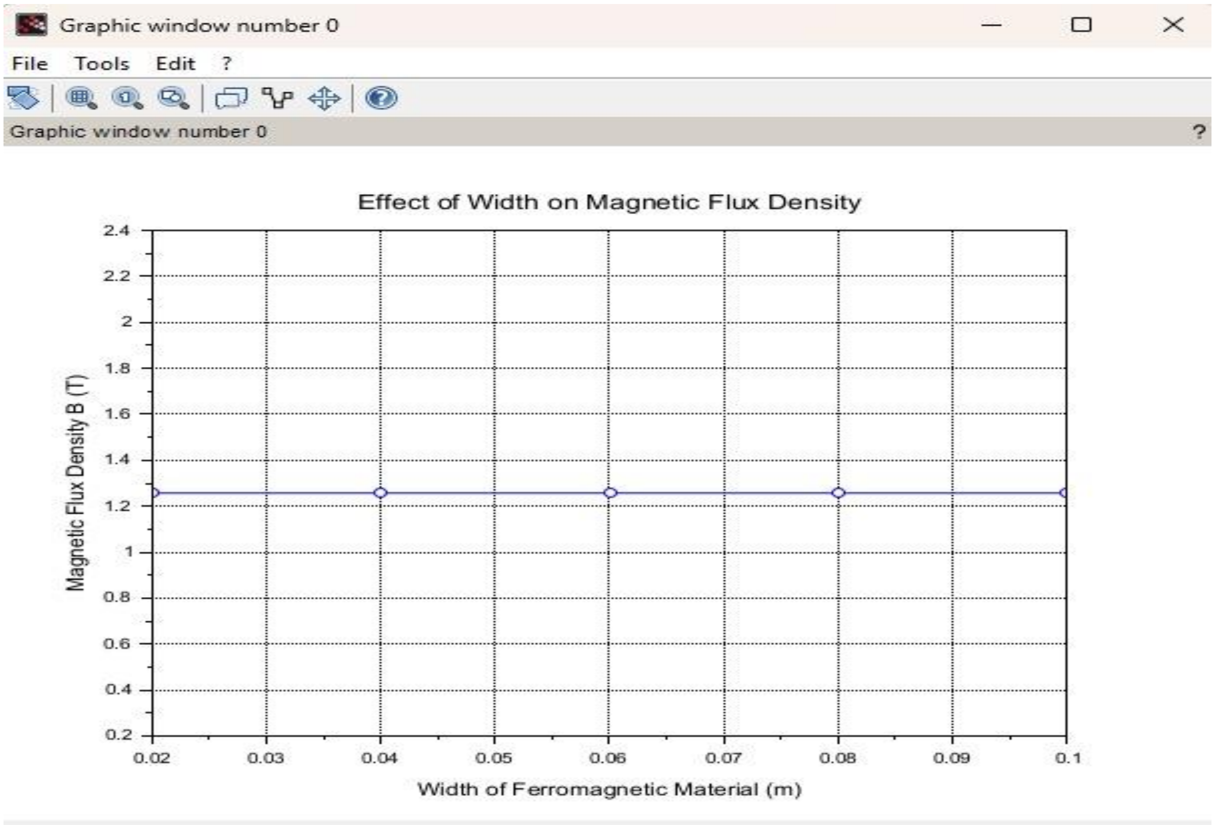


```
clc;
clear;
clf;

// Constants
L = 0.1;                    // Length in meters (10 cm)
H = 1000;                   // Magnetic field intensity (A/m)
mu_0 = 4 * %pi * 10^(-7);   // Permeability of free space (H/m)

// Widths in meters (2 cm to 10 cm)
widths = [0.02, 0.04, 0.06, 0.08, 0.10];
num_widths = length(widths);

// Relative permeability of soft iron
mu_r = 1000;

// Initialise array
B_values = zeros(1, num_widths);

// Calculate magnetic flux density
for i = 1:num_widths
    B_values(i) = mu_0 * mu_r * H;
end

// Display results
disp("Width (m)     Flux Density (T)");

for i = 1:num_widths
    mprintf("%.2f          %.6f\n", widths(i), B_values(i));
end

// Plot the results
plot(widths, B_values, '-o');

xlabel("Width of Ferromagnetic Material (m)");
ylabel("Magnetic Flux Density B (T)");
title("Effect of Width on Magnetic Flux Density");

xgrid();
```

Output:



```
 "Width (m)     Flux Density (T)"
0.02          1.256637
0.04          1.256637
0.06          1.256637
0.08          1.256637
0.10          1.256637

-->
```



Case 3: To examine how the magnetic flux density (B) in a ferromagnetic material changes with different values of magnetic field intensity at constant length and width.

Conditions:

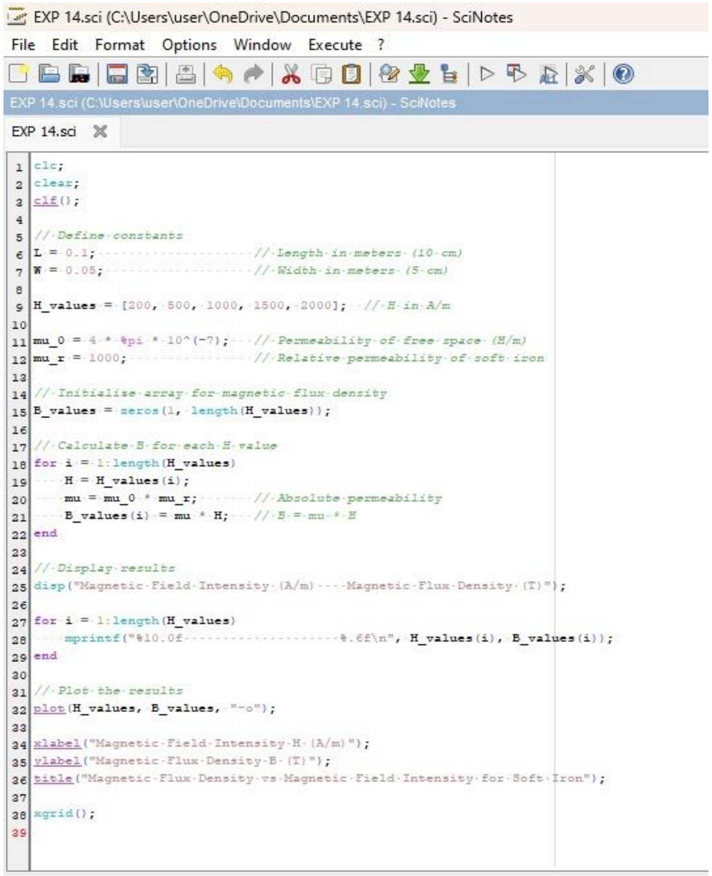
● Material: Soft iron (same as in the previous studies)

● Length (L): Constant at 10 cm

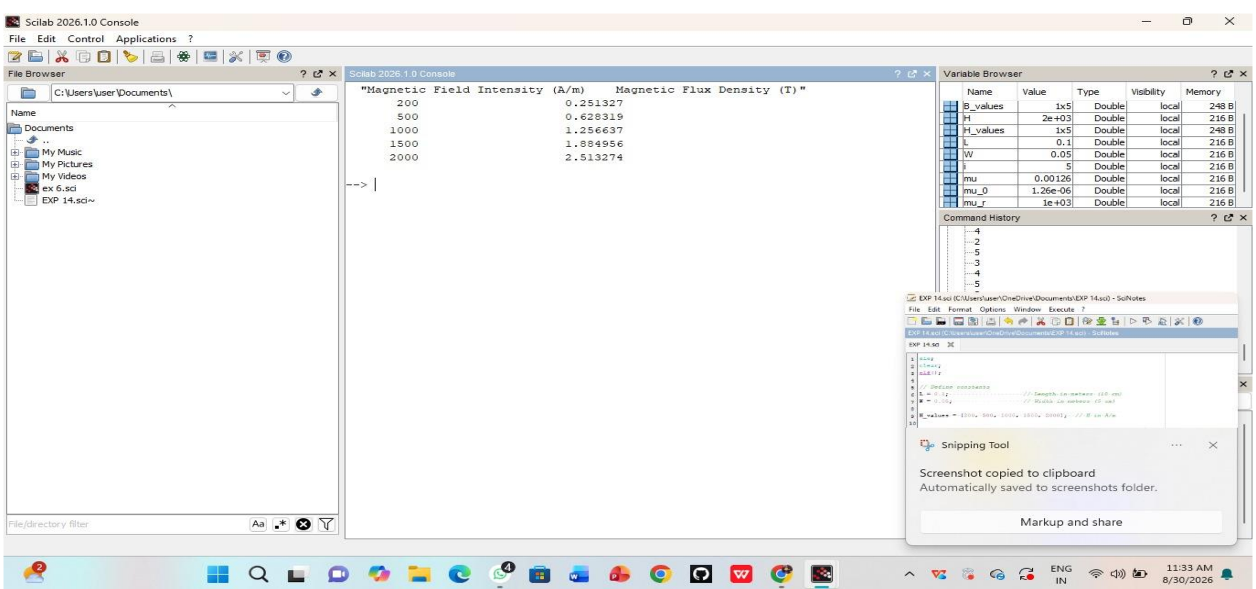
● Width (W): Constant at 5 cm

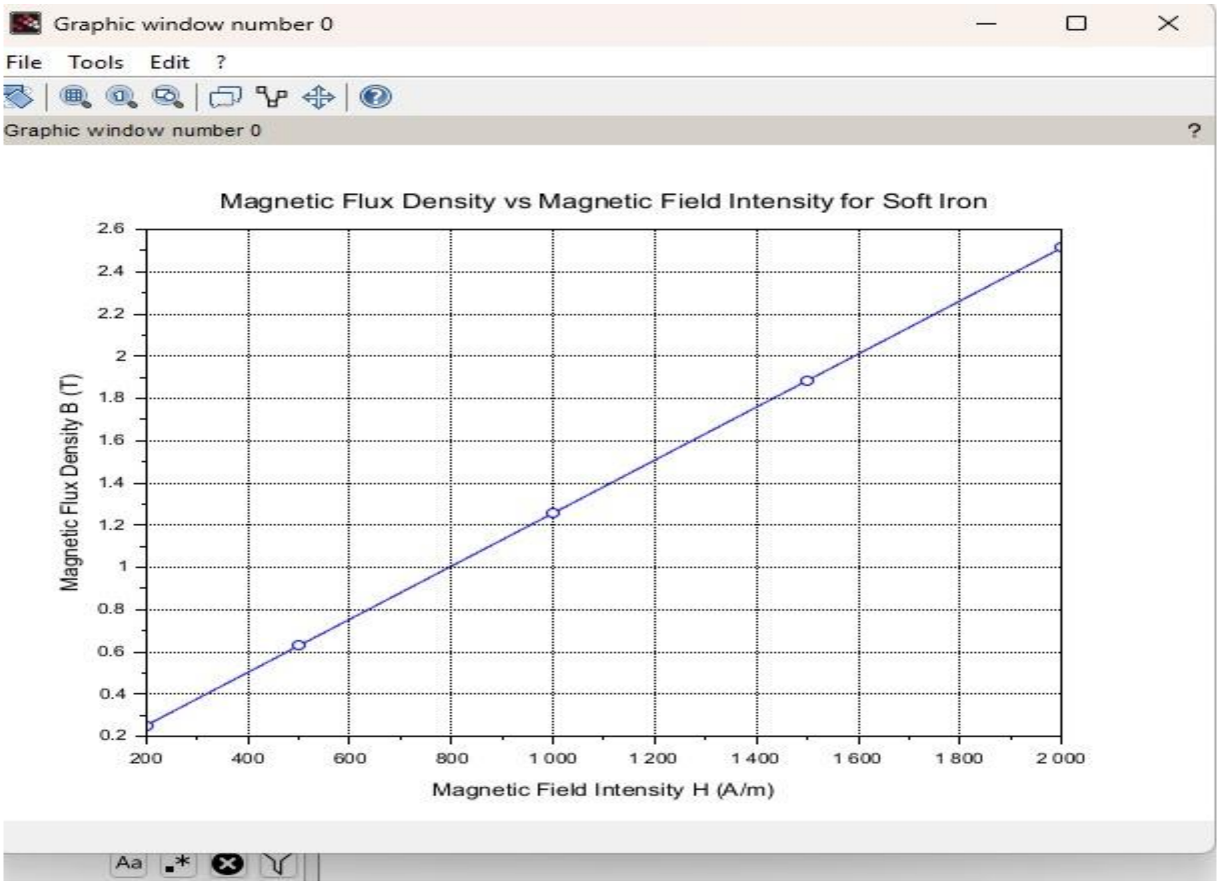
● Magnetic Field Intensity (H): Varies from 200 A/m to 2000 A/m (i.e., 200 A/m, 500 A/m, 1000 A/m, 1500 A/m, 2000 A/m)

Program:



Output:





Result : Thus the program was verified for program for Magnetic flux density in ferromagnetic materials using SCILAB was successfully.